Journal of Photonics for Energy

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Eric S. Muckley, Nicholas Miller, Christopher B. Jacobs, Thomas Gredig, Ilia N. Ivanov, "Morphology-defined interaction of copper phthalocyanine with O_2/H_2O," *J. Photon. Energy* **6**(4), 045501 (2016), doi: 10.1117/1.JPE.6.045501.

Morphology-defined interaction of copper phthalocyanine with O_2/H_2O

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Abstract. Copper phthalocyanine (CuPc) is an important hole transport layer for organic photovoltaics (OPVs), but interaction with ambient gas/vapor may lead to changes in its electronic properties and limit OPV device lifetimes. CuPc films of thickness 25 and 100 nm were grown by thermal sublimation at 25°C, 150°C, and 250°C in order to vary morphology. We measured electrical resistance and film mass *in situ* during exposure to controlled pulses of O_2 and H_2O vapor. CuPc films deposited at 250°C showed a factor of 5 higher uptake of O_2 as detected by a quartz crystal microbalance (QCM), possibly due to the formation of β-CuPc at $T > 200$°C which allows higher O_2 mobility between stacked molecules. While weight-based measurements stabilize after ~ 10 min of gas exposure, resistance response stabilizes over times > 1 h, suggesting that mass change occurs by rapid adsorption at active surface sites whereas resistive response is dominated by slow diffusion of adsorbates into the bulk film. The 25 nm films exhibit higher resistive response than 100 nm films after an hour of O_2/H_2O exposure due to fast analyte diffusion down to the film/electrode interface. We found evidence of decoupling of CuPc from the gold-coated QCM crystal due to preferential adsorption of O_2/H_2O molecules on gold. © *2016 Society of Photo-Optical Instrumentation Engineers (SPIE)* [DOI: 10.1117/1.JPE.6.045501]

Keywords: copper phthalocyanine; morphology; oxygen; water; quartz crystal microbalance; adsorption.

Paper 16089P received Aug. 16, 2016; accepted for publication Oct. 14, 2016; published online Nov. 1, 2016.

1 Introduction

The emergence of organic electronics highlights the importance of low-cost functional materials for flexible thin film applications.[1] Copper phthalocyanine (CuPc) is an aromatic macrocycle being explored for use in organic photovoltaics (OPVs),[2] organic light-emitting diodes,[3] organic field effect transistors,[4] biosensors,[5] and gas sensors.[6] Its p-type behavior, strong optical absorption in the UV–Vis wavelength range, and low cost due to industrial-scale production ($>60,000$ tons annually[7]) as a blue pigment for inks, plastics, and textiles make CuPc well-suited for applications in low-cost flexible devices.[8,9]

Although CuPc is thermally and chemically stable under ambient conditions,[10] adsorption of gas molecules can facilitate charge transfer, which alters the intrinsic electronic properties of CuPc films,[11] leading to significant performance loses in OPVs. Electronic conductivity of metal phthalocyanines arises primarily from doping by electron-accepting O_2 and H_2O, which increases hole density and perturbs the surface potential by the addition of negatively charged adsorbate molecules to the film surface.[12,13] Oxygen coordination occurs at the central metal atom[14] while H_2O sorption is generally attributed to hydrogen bonding near crystal

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domain boundaries.[15,16] Exposure to higher H_2O partial pressure leads to the formation of H_2O multilayers on CuPc domain surfaces, resulting in switching from electronic to ionic conduction, which further increases conductivity.[17,18] Moreover, O_2/H_2O adsorption may facilitate fracture, displacement, and rearrangement of CuPc domains by disturbing intermolecular and molecule-substrate adhesion.[19]

The response of metal phthalocyanines to organic vapors and noxious gases has been well-characterized.[16,19,20] It is recognized that CuPc film morphology and layer thickness play an important role in the electronic properties and gas response of CuPc films, but the relationship between morphology, mass loading by analyte molecules such as H_2O/O_2, and the electrical response of CuPc has not been thoroughly investigated.[21–23] During thermal deposition of CuPc, molecules align by weak van der Waals and π-π interactions, resulting in electron delocalization along the Cu atom stacking axis.[9–10] Substrate heating enhances mobility of deposited molecules, allowing formation of longer molecular chains that further self-assemble to form elongated crystallites. Crystalline domain growth is often associated with changes in film porosity and roughness.[17,24] Faster gas response can be achieved for higher film porosity due to increased surface area.[15,25] Surface area and the number of intergrain boundaries that impede charge transport typically decrease as domains grow larger.[26] Since response time is primarily related to slow diffusion of adsorbates into the film bulk,[11] the response time is inversely proportional to film thickness, since surface area to volume ratio decreases as thickness increases.[10] In thinner films, charge transport layers lie closer to the film/gas interface, therefore, the effects of adsorption-induced surface trap states are enhanced.[20] An understanding of the influence of film morphology on CuPc gas response is critical for extending device lifetimes and improving gas sensor performance.

Here, we study the CuPc gas/vapor response by measuring changes in electrical resistance and mass of films exposed to O_2 and H_2O. This allows us to directly observe the relationship between gas sorption/desorption kinetics and the electrical response of the same film to gas/vapor exposure.

2 Experiment

Purified CuPc was purchased from Sigma-Aldrich and used without purification. CuPc films were deposited by thermal sublimation onto precleaned platinum interdigitated electrodes and gold-plated 5 MHz AT-cut quartz crystal microbalance (QCM) crystals. Film deposition was carried out in a thermal evaporator (Nanomaster NTE-3000). During thermal evaporation, substrates were held at 25°C, 150°C, and 250°C. Film thickness was monitored *in situ* using a QCM system to produce both 25- and 100-nm thick films at each deposition temperature.

Prior to gas/vapor sorption measurements, films were placed under 10^{-6} Torr (1.33×10^{-6} mbar) vacuum for 3 h. QCM frequency change was recorded using an SRS QCM200 and electrical resistance was recorded using a Keithley 6430 source-meter with pre-amplifier. Pulses of O_2 and H_2O vapor with pressures 5, 10, 15, and 20 Torr were injected into the vacuum chamber using two mass flow controllers. A LabVIEW program was used to control the gas/vapor flow and record resistivity and QCM frequency shift during the experiment. All measurements were performed at 27 ± 0.5°C using a modified pressure/flow control system from surface measurement systems. Frequency shift of the QCM crystals during the experiment was converted to mass change using the Sauerbrey equation.[27]

To confirm that morphology of the CuPc films changed with deposition temperature, atomic force microscope (AFM) images of each film were acquired using a NTEGRA Spectra microscope (NT-MDT). The AFM was operated in semicontact (tapping) mode at 73 kHz tip resonance frequency with a 0.7 Hz line scan rate.

3 Results and Discussion

AFM images of 25-nm (top row) and 100-nm thick (bottom row) CuPc films deposited at 25°C, 150°C, and 250°C are shown in Fig. 1. Film morphology is consistent with temperature-induced grain size growth observed in previous studies on CuPc films.[26] Increasing deposition

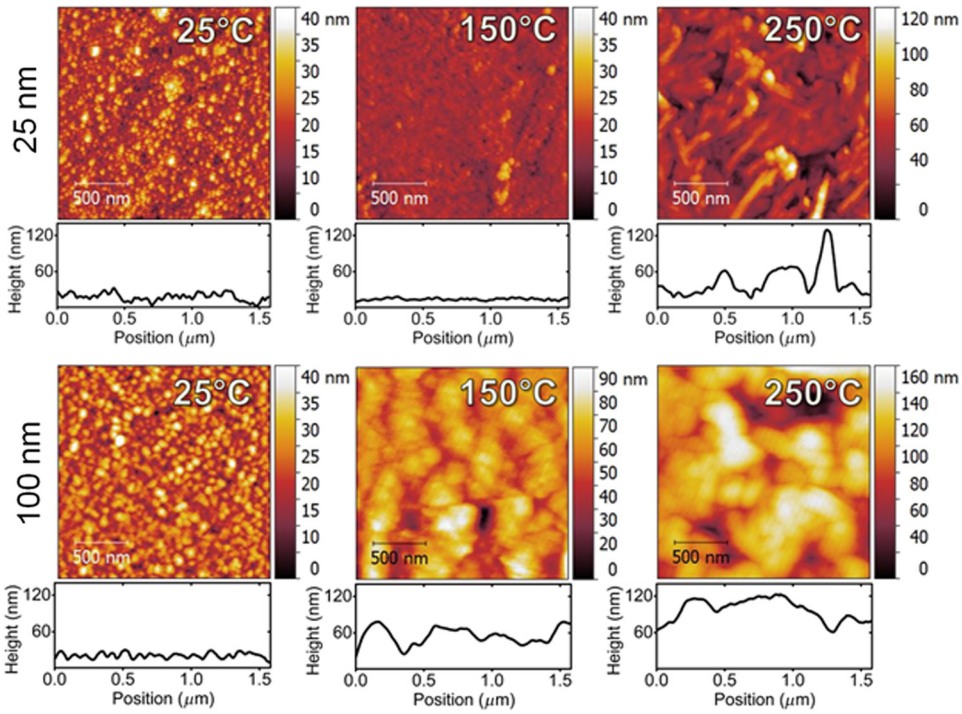

Fig. 1 AFM images of 25- and 100-nm thick (top and bottom rows, respectively) CuPc films deposited on glass substrates at 25°C, 150°C, and 250°C. Line profiles correspond to the bottom row of pixels in each image.

temperature results in larger elongated domains due to increased molecular diffusion leading to the formation of a longer molecular stacking along the axis of the central Cu atom. The 100 nm films generally contain larger domains than the 25 nm CuPc films. In all cases except for the 25 nm film deposited at 150°C, film roughness increases with the deposition temperature. While short-range roughness ($r < x_i$, where x_i is the correlation length or size of the grain) decreases at higher deposition temperatures due to flattening of the domains, long-range roughness ($r > x_i$) increases with temperature, likely due to the presence of structural imperfections such as pinholes which increase film porosity at higher temperatures. Such structural imperfections are speculated to arise due to Ehrlich–Schwoebel-like energy barriers that break the domain growth symmetry.[28]

Resistance response (R/R_0) and mass change (ΔM) of CuPc films exposed to different pressures of H_2O and O_2 are shown in Fig. 2 during 4-pulse sequences of H_2O and O_2 (shown in bottom panel). Resistance of CuPc decreases in response to both H_2O and O_2, indicating that both analytes behave as electron acceptors. H_2O/O_2 sorption leads to charge transfer which increases hole density, resulting in higher conductivity, in agreement with previous studies.[12,13] In films deposited at temperatures over 25°C, resistance change occurs over long time scales (>1 h), even during isobaric conditions. Thus, not all films reach an equilibrated steady-state condition before the next gas pulse begins. During the 5 Torr H_2O pulse, R/R_0 and ΔM kinetics are qualitatively similar to those of the next pulses. This suggests that blocking of active adsorption sites results in surface activity, which is similar to that of pristine surfaces, even though some sites are occupied. Since the focus of this experiment is to study the effect of morphology on gas response, and each film is exposed to the same sequence of gas pulses, the gas/vapor responses were used to compare between different films without reaching equilibrium.

As shown in the central panel of Fig. 2, mass loading of films changes most appreciably during O_2 exposure and is proportional to O_2 pressure. The response dynamics were further examined using background-corrected resistance ($R/R_0 = 1$) and mass normalized to $\Delta M = 0$ just before exposure to the 20 Torr H_2O/O_2 pulses. The results are summarized in Figs. 3(a) and 3(b) for H_2O and O_2 response of the CuPc films. Vertical dashed lines in Figs. 3(a) and 3(b) identify the beginning and end of gas/vapor exposure. Corresponding

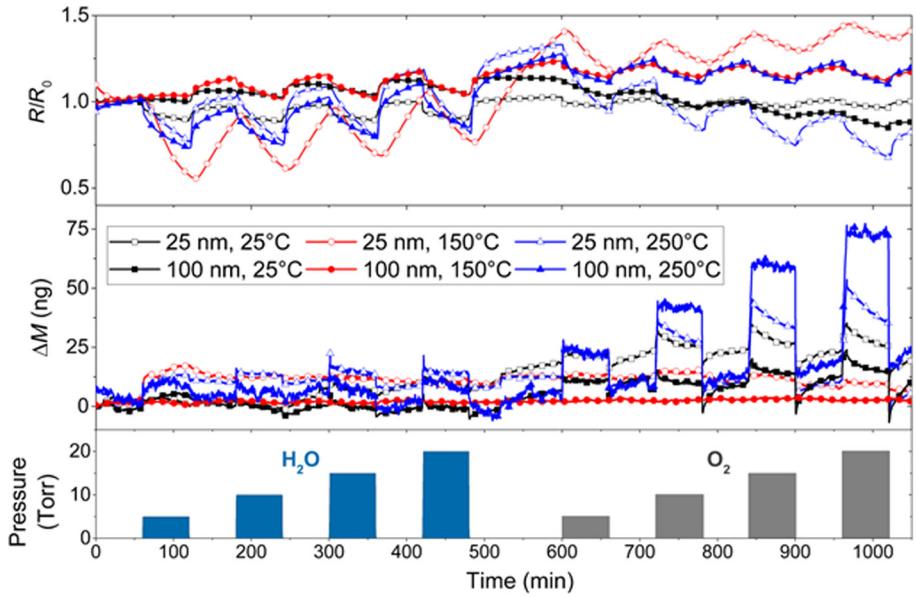

Fig. 2 (top panel) Resistance response (R/R_0) and (middle panel) mass change (ΔM) of CuPc films during exposure to (bottom panel) different pressures of H_2O vapor (blue colored pulses) and O_2 (gray colored pulses). Films are labeled by their thickness (25 or 100 nm) and the temperature of the substrate during deposition (25°C, 150°C, or 250°C), with a generic name of *XX* nm, *YY*°C.

changes in mass during the gas/vapor exposure are shown in Figs. 3(c) and 3(d). Upon initial exposure to H_2O/O_2, rapid mass loading occurs, primarily in films deposited at 250°C. The change in mass ΔM is larger for 100 nm films than 25 nm films, which is expected since thicker films provide more active adsorption sites than the thinner films. Film recovery (desorption of H_2O/O_2) occurs over <5 min after exposure of films to vacuum. Mass change in the CuPc appears almost instantaneous and is completely reversible. In contrast, changes in resistance

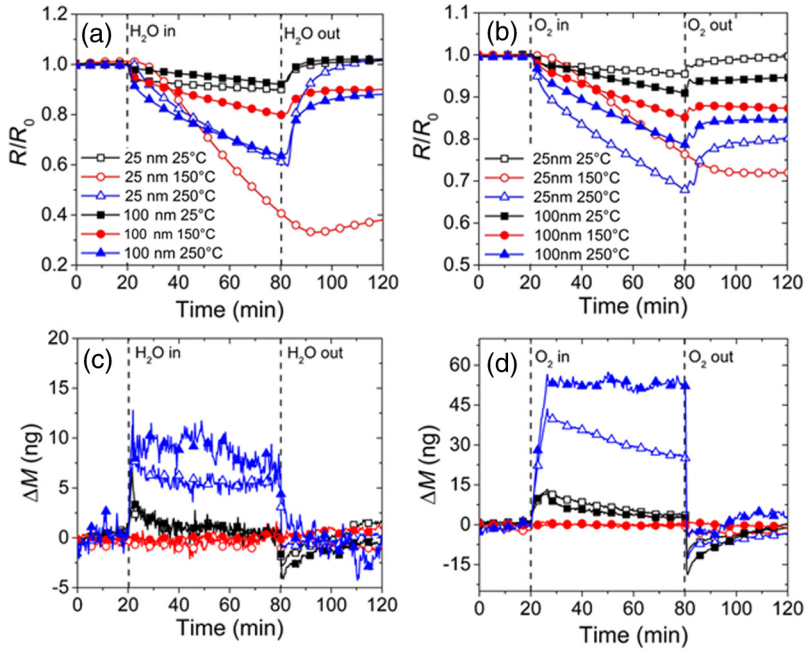

Fig. 3 Normalized resistance change (R/R_0) during exposure to 20 Torr pulses of (a) H_2O vapor and (b) O_2 for 1 h. Normalized mass change (ΔM) in response to exposure of CuPc films to 20 Torr (c) H_2O vapor and (d) O_2 for 1 h.

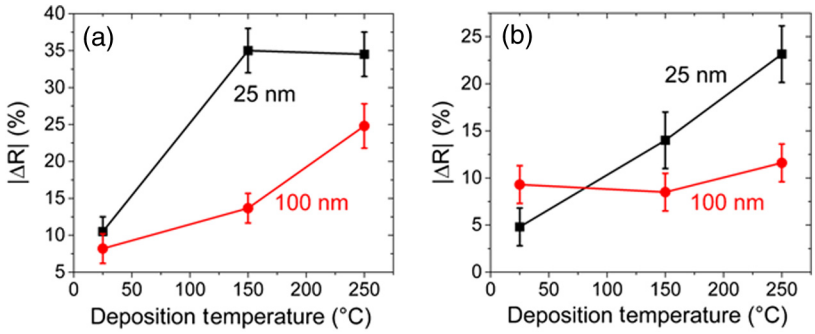

Fig. 4 The resistance change ($|\Delta R|$) of 25- and 100-nm thick films during 1 h exposure to 20 Torr pulses of (a) H_2O vapor and (b) O_2 as a function of film deposition temperature.

of CuPc occur over much longer time scales (>1 h) and are not completely reversible, especially for O_2 desorption. The difference in R/R_0 and ΔM response times suggests that different mechanisms are responsible for resistance and mass changes of CuPc films. Mass loading of CuPc films and at domain boundaries occurs rapidly but does not significantly alter charge carrier density near the charge transport layers (located at the film–substrate interface). Since films were deposited on substrates with prepatterned electrodes, it is likely that only charge carriers in domains near the film–substrate interface contribute to the resistive gas/vapor resistance. Only after adsorbates diffuse deep enough into the bulk film to reach charge transport regions near the CuPc–electrode interface, which occurs on the order of hours, does the conductivity increase. Kerp et al.[18] reports that mass loading saturation of micron-thick zinc phthalocyanine films by O_2 adsorption occurs on a time scale of minutes. The data reported here suggest that the saturation reported by Kerp et al.[18] is a consequence of adsorbate crowding at surface adsorption sites, not saturation of the bulk film by O_2, which occurs over hours. The slow diffusion of adsorbates toward the film–electrode interface observed here is consistent with the findings of Passard et al.[11] who found that gas response occurred over time scales >1 h in 300 nm polyfluoroaluminum phthalocyanine films.

It should be noted that the 25 nm film deposited at 150°C exhibits the highest resistive H_2O response and the slowest response kinetics. The initial resistive response does not take place until after 5 min after the gas/vapor pressure in the measurement chamber has changed. This is likely related to the morphology of the film, as shown in Fig. 1. The 25 nm film deposited at 150°C exhibits the lowest roughness of any other film in the series, which suggests a high density and low film porosity. The low porosity leads to slow adsorbate diffusion in and out of the film, resulting in slower response kinetics and poor reversibility. This film also appears to have a lower average thickness than the other films and subsequently a higher eventual resistive response.

Figure 4 shows magnitude of the resistance change (ΔR) of each film after 1 h exposure to 20 Torr pulses of H_2O vapor (a) and O_2 (b). The ΔR response to both H_2O and O_2 generally increases with film deposition temperature. This is likely due to increases in film porosity, which results from the growth of larger domains. The porosity improves access to active adsorption sites by facilitating adsorbate diffusion through open channels, increasing the rate at which adsorbates reach charge transport layers near the film/electrode interface. ΔR is generally higher for 25 nm films than 100 nm films since adsorbate migration to electrodes occurs more rapidly. This finding agrees with the results of others who show that film thickness is directly related to the magnitude and response time of the resistive CuPc gas/vapor response.[11,25]

Mass changes in films during 20 Torr gas/vapor pulses are shown in Fig. 5(a). As mentioned previously, mass loading occurs most significantly in films deposited at 250°C. Mass loading of O_2 is enhanced over a factor of 5 in β-phase CuPc films deposited at 250°C. At roughly 200°C, CuPc undergoes a metastable triclinic α-phase to stable monoclinic β-phase transition characterized by an increase in the tilting angle between the molecular and stacking axes, resulting in less π-π overlap between adjacent molecules.[10] The transition increases spacing between the central metal atoms of adjacent stacked molecules from roughly 3.8 to 5 Å.[29] Since the van der Waals radius of O_2 is approximately 1.5 Å,[30] extra space between β-phase stacked molecules

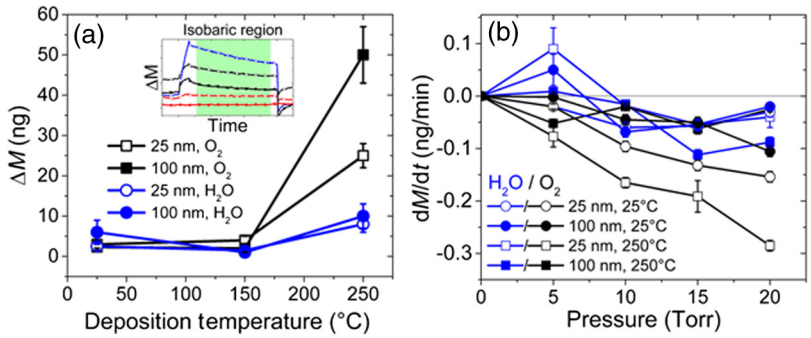

Fig. 5 (a) Total mass change (ΔM) in response to 20 Torr pulses of O_2 (black) and H_2O (blue), shown for 25- and 100-nm thick films deposited at 25°C, 150°C, and 250°C. The inset is related to (b). (b) Average time derivative of film mass (dM/dt) during O_2 exposure (black) and H_2O exposure (blue) after initial rapid mass loading has occurred. dM/dt is extracted during O_2/H_2O exposure under isobaric conditions, highlighted in green in the inset of (a). For clarity, data from films deposited at 150°C are not shown since $dM/dt \approx 0$ g/s.

may allow greater O_2 mobility toward central Cu atoms, so the formation of β-phase domains may significantly enhance O_2 coordination and increase the amount of physisorbed O_2 in CuPc films deposited at 250°C.[14] The CuPc $\alpha \rightarrow \beta$ transition is also associated with an increase in film roughness,[31] which leads to higher surface area and an increase in the number of accessible adsorption sites. This effect may be responsible for the slight increase in H_2O loading of films deposited at 250°C. Since H_2O adsorption occurs primarily around grain boundaries,[15,16] it is expected that the $\alpha \rightarrow \beta$ transition has less of an effect on H_2O loading than on O_2 loading. The 100-nm thick films exhibit higher ΔM than 25-nm thick films because they necessarily contain more adsorption sites and thus allow accommodation of more adsorbate molecules.

It is informative to examine the behavior of ΔM during H_2O/O_2 exposure. The average time derivative of film mass (dM/dt) was extracted from ΔM curves shown in Fig. 2 during H_2O/O_2 exposure. dM/dt values were averaged over isobaric conditions [highlighted in green in the inset of Fig. 5(a)]. Continuous H_2O/O_2 loading of rigid films would result in a positive value for the derivative ($dM/dt > 0$) during the entire length of the gas/vapor pulse. However, after an initial mass increase, which occurs on a time scale of ~5 min, we observe that the sign of the derivative changes ($dM/dt < 0$) in all films except for the 5 Torr H_2O pulses. Since, it is unlikely that significant desorption of adsorbates occurs during isobaric conditions, we propose that sorption-induced changes in mechanostructural and/or viscoelastic properties of the films are responsible for the apparent decreases in mass. Voinova et al.[32] showed that materials undergoing changes in viscoelasticity may display signatures of mass loss during QCM measurements. The Sauerbrey equation, which is strictly valid only for thin uniform films which are rigidly bound to the quartz oscillator, is no longer valid.[27] The apparent loss of mass has also been observed in other organic films undergoing adsorption of H_2O and O_2 on QCM crystals.[33] Since H_2O sorption generally occurs by hydrogen bonding near grain boundaries[15,16] and can result in the formation of an H_2O meniscus on the CuPc surface,[17,18] it is likely that high H_2O pressures leads to the formation of H_2O multilayer structures near intergrain interfaces. This increases the probability of slippage between adjacent grains as they oscillate at the 5 MHz fundamental frequency of the QCM. Intergrain and CuPc-gold slip motion increase the resonant frequency of the oscillator, which results in the apparent decrease in film mass. Ojifinni et al.[34] reported that H_2O adsorption on gold surfaces may result in the formation of H_2O—Au, H_2O—O, or OH complexes in the presence of oxygen adatoms. Moreover, evidence of CuPc molecule detachment/rearrangement on gold substrates due to preferential O_2/H_2O adsorption has been reported.[19] Both of these processes represent possible pathways for decoupling of rigid CuPc–gold interactions which increases f_0 as CuPc molecules are effectively lifted off the QCM surface or their network is disrupted. The positive value of dM/dt during H_2O loading at 5 Torr provides evidence that the limited water adsorption at this pressure does not induce significant mechanostructural changes in the films.

The mass rate dM/dt exhibits higher pressure dependence during O_2 loading than during H_2O loading. O_2 coordination with central Cu atoms[14] may disrupt π-π stacking and lead to domain fracture and rearrangement of CuPc.[19] Preferential adsorption of O_2 on gold may be responsible for displacing CuPc from the QCM surface, leading to an increase in f_0. The O_2- induced decrease in mass is most evident in 25-nm thick films, which show a roughly linear decrease in the value of the derivative dM/dt with O_2 pressure. Slow diffusion of O_2 through the CuPc[11] to the QCM surface must occur before disruption of CuPc-gold adhesion takes place, so it is reasonable that dM/dt observed in 25 nm films is more pronounced than in 100 nm films. The pressure dependence of dM/dt is a result of faster diffusion of O_2 into the bulk film and to the film–crystal interface when more adsorbate molecules are present.

4 Conclusions

We investigated separately the effects of film morphology and thickness on the interaction of H_2O and O_2 with CuPc. By measuring electrical resistance of films and frequency shift of CuPc-coated QCM crystals, we studied the relation between sorption/desorption kinetics and film conductivity. While mass loading occurred relatively quickly after gas/vapor exposure (<10 min), resistance changes occurred over much longer time scales (>1 h). This suggests that the resistive CuPc gas/vapor response is dominated by slow diffusion of adsorbates into the film bulk, while mass change occurs by rapid adsorption at active surface sites. Resistive response to analyte gas/vapor generally increases with film deposition temperature due to increased porosity stemming from larger crystal domains. A 25-nm thick films exhibit higher resistive response than 100-nm thick films due to the smaller analyte diffusion length required for reaching the film/electrode interface. O_2 loading of films was enhanced by a factor of 5 in films deposited at 250°C. We suggest that this may be a consequence of the CuPc $\alpha \rightarrow \beta$-phase transition which occurs around 200°C and results in higher O_2 mobility between stacked molecules, facilitating O_2 coordination at the central Cu atoms. Finally, we found evidence of H_2O/O_2-induced mechanostructural changes in the films. This is likely due to slipping motion between CuPc crystallites, and between CuPc and the Au electrode of the QCM oscillator. We propose that mechanical decoupling of CuPc from the gold-coated QCM crystal is caused by preferential adsorption of H_2O/O_2 molecules on gold.

Acknowledgments

This research was conducted at the Center for Nanophase Materials Sciences, which is a DOE Office of Science User Facility. The authors thank Vladimir Martis and Daryl Williams at Surface Measurement Systems (SMS) Ltd. (London, UK) for their support in development of new environmental testing capabilities.

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